

Luka Pirroni

Mechanical Engineering

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SUMMARY:

Results-driven Mechanical Engineering (Automotive) graduate, with strong technical knowledge in CAD, FEA and engineering principles developed through Formula Student and a range of design projects. Demonstrated ability to lead technical teams, apply critical thinking to complex design problems, and deliver innovative, cost-conscious solutions from concept through to validated design under high-pressure deadlines.

EXPERIENCE

INTAKE SYSTEM DESIGNER

Kingston Racing FS Team (09/2025 - 08/2026)

- Designed and repositioned the intake to increase plenum volume for improved charged air capacity while satisfying Formula Student packaging constraints.
- Developed full SolidWorks CAD assembly model, conducting hand calculations for plenum size and runner length, and ensuring compliance with FS regulations.
- Developed in Coordination with Aerodynamics and Powertrain teams.

Vehicle Dynamics Simulation (09/2025 - 08/2026)

- Developed a conservative baseline vehicle model in IPG CarMaker, iteratively optimising setup parameters across FS dynamic events, and reducing lap times by 31% while maintaining valid competition track boundaries.

BRAKE PEDAL STRUCTURAL OPTIMISATION

Kingston University (01/2026 - 03/2026)

- Designed and structurally optimised a Formula Student brake pedal assembly in SolidWorks and ANSYS Mechanical, conducting hand calculations and FEA across multiple load cases.
- Achieved over 30% mass reduction on two critical components while maintaining a safety factor of 1.97, satisfying all DFM/DFA constraints for CNC machining and Formula Student regulations.

PORTABLE GANTRY CRANE DESIGN

Kingston University (12/2025 - 04/2026)

- Led a team of 8 to design, analyse and validate a portable gantry crane in SolidWorks and ANSYS Mechanical, delegating tasks based on individual strengths and managing project deadlines.
- Personally delivered the core CAD and FEA across critical components including the I-Beam, gears, shaft and trolley, validating results against hand calculations using Shigley methods.
- Confirmed structural integrity across all components with a minimum safety factor of 4.4, while optimising for portability, ease of assembly and parts cost through material selection and design decisions.

BIO-INSPIRED CRASH STRUCTURE

Individual Project | Kingston University (09/2025 - 04/2026)

- Designed, simulated and experimentally validated a gyroid TPMS lattice against a conventional hexagonal prism using SolidWorks, nTop, and ANSYS Explicit Dynamics.
- Achieved the highest crush force efficiency (0.893) across all tested specimens, demonstrating superior isotropy under lateral loading and identifying the gyroid as the most promising Formula Student impact attenuator candidate.

Education

BEng (Hons) Mechanical Engineering (Automotive)
1st Class Honours

Kingston University
(09/2023 - 07/2026)
Kingston Upon Thames, UK

TECHNICAL SKILLS

- CAD
- FEA
- CFD
- Structural Optimisation
- Vehicle Dynamics Simulation
- DFM/DFA
- First Principles Analysis
- FDM 3D Printing

Software

- SolidWorks (CSWP Certified)
- ANSYS Mechanical
- IPG CarMaker
- nTop
- Microsoft Office

Academic Experience:

- Python
- Matlab & Simulink

KEY ACHIEVEMENTS

- Best Presentation Award: Achieved best presentation award presenting a child safety seat design in collaboration with Jaguar/LandRover.
- Team Lead Success: Every team project I have led in Kingston University, achieved A grade.

Languages

English: Native
French: Proficient